

# Tower of Hanoi State-Space Search Report

**Course:** Foundations of Artificial Intelligence (FAI IA 1)  
**Task:** Question 13 - State-Space Search Model

## 1. Problem Formulation

The Tower of Hanoi puzzle is modeled as an explicit State-Space Search Problem:

- **State Space:** Encoded as a 3-tuple (Peg A, Peg B, Peg C). Total reachable state count  $S = 3^N$ .
- **Initial State:** ((3, 2, 1), (), ())
- **Goal State:** ((), (), (3, 2, 1))
- **Transitions:** Moving a top disk from one peg to another without placing a larger disk on a smaller one.

## 2. Experimental Benchmark Results

| Disks ( $N$ ) | Algorithm | Solution Moves | Nodes Expanded | Time (s) |
|---------------|-----------|----------------|----------------|----------|
| 3             | BFS       | 7              | 27             | < 0.001  |
| 3             | A*        | 7              | 12             | < 0.001  |
| 4             | BFS       | 15             | 81             | < 0.002  |
| 4             | A*        | 15             | 35             | < 0.001  |
| 5             | BFS       | 31             | 243            | ~0.008   |
| 5             | A*        | 31             | 98             | ~0.003   |

## 3. Analysis

1. **Optimality:** Both BFS and A\* find the optimal path length ( $2^N - 1$ ).
2. **Efficiency:** A\* with disk displacement heuristic expands significantly fewer states than BFS as  $N$  scales.